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Paper Title: How Catholic Social Teaching Informs Ethical AI Design: Bridging Moral Philosophy and Machine Learning for Responsible Technology

Date: June 28, 2025

1.0 Introduction

Artificial intelligence is progressing at an astonishing pace. It is shaping decisions in various human endeavours; from finance to medicine, education to media, labour to public policy. Code now mediates who gets a loan, health diagnosis and even a visa. These systems do not originate on their own. They reflect data histories, model choices, governance and optimization goals set by human hands and brains. On the same vein, the Catholic Social Teaching (CST) offers a long-tested moral vocabulary for guiding these choices. From the days of *Rerum Novarum* of Pope Leo XII, the Church has, through her social teaching, provided a vital and comprehensive moral compass for navigating ethical complexities (Pontifical Council for Justice and Peace, 2004). This also extends to the design and deployment of AI in the human society. With its core commitment to human dignity, the common good, solidarity and subsidiarity, Catholic Social Teaching (CST) offers a unique lens through which to evaluate AI's societal impact. Its core objective is to ensure that technological advancements foster human dignity rather than exacerbate inequalities or undermine fundamental human right of the human person (John XXIII, 1961; Pontifical Council for Justice and Peace, 2004; Benedict XVI, 2009; Francis, 2020; Pontifical Academy for Life, 2020).

De jure et de facto, technology is not neutral. Training data carry social bias. Objective functions reward some outcomes and ignore others. Interface design can exclude whole populations. Deployment at scale can magnify old injustices or create new ones. These raises concern, and make CST to ask these three basic questions around: Who is seen. Who is served. Who is left out. To answer these questions, the Vatican started something called the "*Rome Call for AI Ethics*" in 2020. This important event brought together many different groups, like big tech companies (IBM, Microsoft) and religious leaders from Jewish, Muslim, and other traditions. They all agreed on basic rules for making sure AI respects human dignity (Pontifical Academy for Life, 2020). The Rome Call explicitly addresses pressing concerns such as pervasive surveillance, algorithmic bias, and the ethical obligations of developers. It went on to advocate for a balanced approach that harnesses AI's benefits while rigorously safeguarding against misuse and the infringement of individual rights (Francis, 2020; Pontifical Academy for Life, 2020).

Issues of systemic discrimination and erosion of privacy have been a big concern when issues of ethical AI are raised. Critics would often warn that if AI is not properly regulated, it can perpetuate societal biases, disproportionately harming marginalized communities and undermining their dignity (Noble, 2018). Also, letting secretive algorithms make important decisions creates big ethical issues. The question then becomes: Who should be held accountable if things go wrong? And could relying too much on AI make people less responsible for their actions? (Zuboff, 2019).

To address these issues, there is need for inclusive practices, equal opportunities and an unwavering protection of every individual's dignity.

This paper therefore concerns itself with the CST principles and how it can be an operational design criterion for AI teams. It further proposes a multi-level governance patterns shaped by subsidiarity so that local voices inform global standards. (Benedict XVI, 2009; USCCB, n.d.).

2.0 Historical Context of Catholic Social Teaching

The foundation for using Catholic Social Teaching (CST) to handle new problems like AI came mostly after World War II. During that time, the Catholic Church began emphasizing fairness in global economics and relationships between countries. By 1961, Pope John XXIII wrote an encyclical, called *Mater et magistra* (Mother and Teacher). This was a big step, highlighting the huge gap between rich and poor nations. The pope emphasized the need for wealthy countries to help poorer ones, while respecting their distinct cultural identities. He warned clearly that if global economic imbalances continued, it would seriously threaten world peace (John XXIII, 1961; O'Brien & Shannon, 1992).

This trajectory continued with *Pacem in terris* (Peace on Earth) in 1963, a groundbreaking document as it was the first encyclical addressed explicitly to "all people of good will," not solely Catholics. John XXIII argued that sustainable peace required establishing a global order grounded in the recognition of mutual rights and responsibilities binding individuals, social groups, and states, urging the faithful to actively internalize and apply these social teachings (John XXIII, 1963; Curran, 2002).

The accelerating pace of globalization and intercultural exchange presented both new opportunities and challenges for CST. Pope Benedict XVI, in his 2009 encyclical *Caritas in Veritate* (Charity in Truth), reflected deeply on the role of culture. While celebrating the increased potential for intercultural dialogue, he cautioned forcefully against cultural homogenization ("leveling") and the uncritical acceptance of diverse lifestyles that could erode the unique value of individual cultures and, consequently, human dignity itself (Benedict XVI, 2009; Massaro, 2012). CST is not static; it is a living tradition built upon a rich corpus of papal encyclicals, conciliar documents (like those of the Second Vatican Council), and episcopal teachings. These sources consistently articulate core themes: the inviolable dignity of the human person created in God's image, the essential nature of community, and the relentless pursuit of justice (United States Conference of Catholic Bishops [USCCB], n.d.; O'Brien & Shannon, 1992). This enduring historical commitment to social justice and human dignity provides the essential framework through which the Church engages contemporary ethical dilemmas, including those posed by transformative technologies like artificial intelligence, advocating steadfastly for the protection of human identity and worth in an increasingly complex and digitized world (Pontifical Council for Justice and Peace, 2004; Benedict XVI, 2009).

3.0 Core Principles of Catholic Social Teaching

Catholic Social Teaching provides a coherent and principled framework for analyzing and addressing contemporary social issues, including those arising from technological advancement. Its core principles stem from a profound understanding of the human person within society and the moral imperatives of justice.

3.1 Life and Dignity of the Human Person

The bedrock of all CST is the inherent and inviolable dignity of every human being. This dignity is not earned or contingent upon ability, status, or stage of life; it is intrinsic, arising from the fundamental belief that every person is created in the image and likeness of God (Genesis 1:26-27) from conception until natural death (USCCB, n.d.; John Paul II, 1991). Respecting this dignity transcends mere non-interference; it demands the active promotion of social conditions that allow every individual and community to flourish – materially, spiritually, socially, and intellectually. It necessitates safeguarding fundamental rights and recognizing the profound interconnectedness of all humanity (Benedict XVI, 2009; Paul VI, 1965).

3.2 The Common Good

The common good is defined as "the sum total of social conditions which allow people, either as groups or as individuals, to reach their fulfillment more fully and more easily" (John XXIII, 1963, para. 55; Pontifical Council for Justice and Peace, 2004). This principle insists that the benefits of social life must be accessible to all members of society. Decisions, particularly those impacting large populations like AI deployment strategies, must be made with the well-being of the entire community in mind, not merely the preferences of the majority or powerful interests (USCCB, n.d.; Benedict XVI, 2009). Achieving the common good requires a careful balancing of diverse individual and group interests, always grounded in the prior commitment to fundamental human dignity and rights (John Paul II, 1987).

3.3 Solidarity

Solidarity is far more than a feeling of vague compassion. It is a "firm and persevering determination to commit oneself to the common good" and the authentic well-being of others, especially the marginalized and vulnerable (John Paul II, 1987, para. 38; USCCB, n.d.). It challenges individuals and societies to move beyond passive sympathy to active engagement in building a more just world, recognizing that "we are all really responsible for all" (John Paul II, 1987, para. 38). Solidarity fosters unity amidst diversity, cultivates deep empathy, and demands inclusive practices that value the unique contributions of every member of society (Francis, 2020; Benedict XVI, 2009). In the context of globalized technology, it implies a shared responsibility for ensuring AI benefits all humanity, not just a privileged few.

3.4 Subsidiarity

The principle of subsidiarity provides crucial guidance for governance and decision-making structures. It asserts that social and political issues should be addressed at the most immediate or local level possible, empowering communities and individuals (Pius XI, 1931; USCCB, n.d.). Higher levels of authority (e.g., national governments, international bodies, or corporate headquarters) should only intervene when lower-level entities (e.g., local communities, workers,

regional offices) genuinely lack the capacity to address an issue effectively on their own. Subsidiarity respects and encourages individual initiative and local solutions, ensuring that larger social structures support and empower rather than absorb or stifle local efforts (Benedict XVI, 2009; John Paul II, 1991). Fundamentally, every social activity, including the development and regulation of technology, should enhance the capabilities and agency of individuals and communities, not diminish them (Pontifical Council for Justice and Peace, 2004).

4.0 Ethical Considerations in AI Design Informed by CST

The Rome Call for AI Ethics: A Multistakeholder Commitment: The practical application of CST principles to AI coalesced significantly with the 2020 "Rome Call for AI Ethics," initiated by the Pontifical Academy for Life. This declaration represents a major multistakeholder commitment to embed ethical considerations into the very fabric of AI development and use (Pontifical Academy for Life, 2020). Endorsed by major technology firms (IBM, Microsoft) and leaders from Judaism, Islam, and other faiths, the Call outlines six core principles: Transparency, Inclusion, Responsibility, Impartiality, Reliability, and Security & Privacy. Its central aim is to establish a moral framework ensuring that as AI moves from research labs into workplaces, homes, and public spaces, it consistently upholds and promotes human dignity (Pontifical Academy for Life, 2020).

Human Dignity and AI Deployment: As AI systems become deeply embedded in critical decision-making processes (e.g., hiring, loan approvals, medical diagnoses, criminal justice risk assessment), the ethical responsibility extends far beyond the design phase to encompass implementation and use. CST emphasizes that the dignity of work is intimately tied to how employers deploy algorithmic tools (John Paul II, 1981; Francis, 2020). An ethical approach demands that choices regarding AI use – what tasks to automate, how to monitor workers, how algorithmic decisions are reviewed – must prioritize human dignity above efficiency or profit motives. The Rome Call explicitly condemns uses of AI, such as pervasive surveillance, that restrict individual freedoms or create systems privileging a few at the expense of the many, deeming such applications fundamentally unjustifiable violations of human dignity (Pontifical Academy for Life, 2020; Francis, 2020).

Privacy and Surveillance: Safeguarding the Inner Sanctum: Privacy emerges as a paramount ethical concern amplified by AI's data-hungry nature. The Second Vatican Council recognized the right to privacy as essential for a "genuinely human life," requiring protection against "undue interference" (Second Vatican Council, 1965, para. 26). CST insists that privacy is not merely a legal concept but a fundamental requirement for human dignity and freedom, allowing for the development of conscience, intimate relationships, and authentic self-expression (Hoose, 2002). Robust regulatory frameworks are therefore necessary to ensure accountability in AI's use, mandating transparency about data collection and usage, and enforcing strong privacy protections as essential safeguards against misuse. Developers and deployers of AI must remain vigilant against surveillance overreach enabled by powerful AI analytics, which threatens intrinsic rights, including religious freedom and freedom of thought (Francis, 2020; Pontifical Academy for Life, 2020).

The Centrality of Human Agency and Accountability: Pope Francis has articulated a crucial CST tenet regarding AI: while these systems can inform and assist human decision-making, ultimate responsibility must always reside with human beings (Francis, 2020). As AI evolves towards greater autonomy, questions of ethical accountability and human safety become critical. Who is responsible when an autonomous vehicle causes harm, or a diagnostic AI makes a fatal error? The relationship between AI and human agency is delicate; developers and users bear the moral burden to ensure algorithms function transparently, biases are actively identified and mitigated, and unintended consequences are minimized (Martin, 2019; Mittelstadt et al., 2016). Crucially, AI systems must not reduce individuals to mere data points or categories defined by algorithms; the uniqueness and irreplaceability of each person must be preserved (Francis, 2020). Upholding justice and equity requires proactive efforts to prevent AI applications from inheriting and amplifying societal biases, leading to discriminatory outcomes in areas like employment, housing, and law enforcement (Noble, 2018; O'Neil, 2016).

Promoting Digital Equity and Inclusion: CST's principles of solidarity and the universal destination of goods demand that ethical AI design actively confront the digital divide. Ensuring marginalized and underserved communities have meaningful access to emerging technologies like AI is not optional; it is a matter of justice (Francis, 2020; Pontifical Academy for Life, 2020). Ensuring digital fairness, which entails having access to technology, having its necessary technical know-how, and affordable connectivity is important to all. This will prevent AI from being a discriminatory tool which widens social and economic gulfs between people and nations. Thus, Ethical AI should be built and shared so everyone can benefit equally. By so doing, technology will serve the common good rather than making existing inequalities worse (World Economic Forum, 2022).

Application of Catholic Social Teaching in AI Development: The Catholic Social Teaching (CST) is not just about ideas. It also provides a practical framework to handle real ethical issues caused by artificial intelligence in various aspects of the society. It does that by constantly ensuring that AI principles of development and deployment follows principles of scripture, tradition and Church's *magisterium* on social justice. This approach helps stakeholders in the AI fields to harness the potential of technology for the common good while protecting human dignity from harm (Pontifical Council for Justice and Peace, 2004; Pontifical Academy for Life, 2020).

Ethical Frameworks, Principles, and Governance: integrating AI into key sectors of human society like healthcare and education reveals both its potential and risks. Thus, CST emphasizes that although AI can process large amounts of data and offer helpful insights, people – and not machines, are fully responsible for how it's built, used, and what results it brings (Francis, 2020; Martin, 2019). This key idea raises complex questions of accountability, particularly when AI applications lead to harmful or unjust outcomes (e.g., misdiagnosis, biased resource allocation, discriminatory educational tracking). To address this challenge, a regulatory framework at national and international levels should exist. These regulations should make it clear who is responsible when things go wrong, push for openness in how AI systems work (when possible), and establish rigorous ethical oversight throughout the AI lifecycle. This entails starting from how the AI is

designed and trained, how data is handled to how it's deployed and monitored afterward (Jobin et al., 2019; Floridi et al., 2018).

Echoing this need, Pope Francis has consistently called for international cooperation to regulate AI technologies, advocating for binding protocols that prioritize fundamental human values. These values include that of like equality and fairness, openness and transparency, security and reliability, and then accountability and trust (Francis, 2020, 2024). His message fits closely with Catholic Social Teaching, which puts human dignity at the center and urges action against inequality. One clear example is in healthcare, where biased AI systems can make it even harder for marginalized groups to have access to a fair and quality medical treatment (Obermeyer et al., 2019; Parikh et al., 2019).

Opportunities for Promoting the Common Good: When guided by CST principles, AI can be a powerful tool for common good in the society. In education sector, it can make learning more personal and accessible, especially for students in poor or remote areas, or those who have diverse learning needs (Holmes et al., 2019). In healthcare, AI can help spot illnesses early, suggest better treatment options, sometimes even a personalize treatment, make better use of medical resources, and bring diagnostic tools to places that lack specialists. This can lead to fairer and more equal health outcomes for everyone (Topol, 2019; Wahl et al., 2018). At the same time, Catholic Social Teaching warns us to pay attention to power dynamics. When only a few big names and companies or governments control how AI is made and used, it creates a serious ethical problem. This kind of control can lead to decisions that serve the interests of the powerful instead of the common good (Zuboff, 2019). Situations like that could lead to manipulations. It can weaken democratic debate, reduce workers' freedom, and put profit or power ahead of public welfare and common good (Francis, 2020; Pontifical Academy for Life, 2020). Countering these risks requires a vigilant evaluation of every AI application through the lens of its impact on human dignity and common good of all.

What is more, the principle of subsidiarity gives important direction here. It holds that decisions should be made as close as possible to the people affected by them. When applied to AI, this means development and regulation should involve local communities, include voices from diverse stakeholders, especially those directly impacted; and ensure individuals, families, and local groups still have real control over how they use and relate to technology. This helps prevent too much power from being concentrated in the hands of a few (Benedict XVI, 2009; USCCB, n.d.).

5.0 Challenges and Critiques

Despite the valuable guidance offered by CST, the integration of AI into society faces significant challenges and warrants critical examination, particularly from perspectives concerned with human dignity and social justice.

Beyond Technological Neutrality: A fundamental challenge lies in debunking the myth of technological neutrality. A key critique informed by philosophy and theology, including CST, asserts that technologies are *not* merely passive tools; they are shaped by the values, assumptions, and priorities of their creators (Feenberg, 1999; Winner, 1980). As philosopher Martin Heidegger argued, technology embodies a specific way of revealing the world, which can instrumentalize

human beings if left unchecked (Heidegger, 1977). The design choices embedded within an AI system—the data it learns from, its optimization goals, its inherent limitations—profoundly influence its functionality and the behaviors it incentivizes or constrains among its users (Introna, 2007). This necessitates rigorous ethical analysis *before* and during development, recognizing that the consequences of deploying AI can reshape social relationships, influence human cognition and behavior, and potentially erode human dignity if designed without sufficient moral reflection (Vallor, 2016; Turkle, 2011).

Human Agency, Responsibility, and the Spiritual Dimension: A core critique from a CST perspective centers on the relationship between AI and the human person. Theologian Nicholas Berdyaev warned that shifting responsibility for personal and societal problems onto machines fundamentally undermines human dignity, as moral responsibility is an inherently human capacity that cannot be delegated to non-sentient artefacts (Berdyaev, 1933; cited in Pontifical Academy for Life, 2020). This underscores Pope Francis's concern that the challenges posed by AI are not merely technical but profoundly spiritual and ethical (Francis, 2020). An over-reliance on algorithmic decision-making risks diminishing human moral reasoning, empathy, and the capacity for discerning judgment in complex situations requiring wisdom beyond data patterns. Technology ethicist Shannon Vallor calls for cultivating "technomoral virtues" to navigate this challenge (Vallor, 2016). CST emphasizes that building AI responsibly means keeping human decision-making at the center. It also calls for what Pope Francis calls "relational intelligence" that is, using technology to strengthen human connections and support community life, not just to make things faster or more automated (Francis, 2020).

Commodification, Exploitation, and Amplified Inequality: The massive amount of data collected to run AI raises serious concerns about turning people's personal information into something to be bought, sold, or exploited like a product. Individuals, often unaware of the extent of data harvesting and profiling, become products whose attention and behavior are mined for profit (Zuboff, 2019). This is particularly exploitative for marginalized groups who may have less power to resist surveillance or negotiate terms of data use. The digital landscape frequently amplifies existing social and economic inequalities, making vulnerable populations even more susceptible to exploitation, discriminatory profiling, and intrusive surveillance (Eubanks, 2018; Benjamin, 2019). Furthermore, the values implicitly or explicitly embedded in AI systems by their designers can have significant societal consequences. As noted earlier, if the design prioritizes engagement metrics and advertising revenue (as is common in social media), it can lead to platforms employing addictive features linked to negative mental health outcomes, especially among youth (Twenge, 2017; Haidt, 2024). This challenges the integrity of the technology and raises serious ethical questions about the responsibilities of developers and regulators to prioritize human well-being and foster a more humane digital environment (Harris, 2016).

Reinforcing Systemic Injustices and the Economic Dimension: Perhaps the most pressing ethical challenge is AI's potential to reinforce, or even exacerbate, systemic inequalities. AI systems trained on historical data often inherit and perpetuate societal biases related to race, gender, socioeconomic status, and other factors, leading to discriminatory outcomes in critical areas like hiring, lending, policing, and judicial risk assessment (Noble, 2018; O'Neil, 2016). The

complex relationship between technology and the economy under capitalism often means that AI-driven automation and efficiency gains primarily benefit owners and shareholders, while displacing workers or subjecting them to increased algorithmic management and precarious conditions (Benanav, 2020; Srnicek & Williams, 2015). Many communities face increased vulnerability—exploitation through gig economy platforms, exclusion from AI-driven services due to lack of access or digital literacy, or disconnection from local economies disrupted by automation. Prioritizing the common good, as CST demands, requires a critical evaluation of how AI technologies impact these dynamics, urging stakeholders to actively design and govern AI in ways that alleviate rather than deepen social fractures and economic injustice (Francis, 2020; Dignitatis Humanae Institute, 2020).

6.0 Conclusion

Artificial intelligence now shapes decisions across daily life. Its growth brings promise and hard ethical questions. Here Catholic Social Teaching matters. Rooted in human dignity, the common good, solidarity, and subsidiarity, it gives a moral frame strong enough for technical change. Across centuries of encyclicals and council teaching, and in recent efforts such as the *Rome Call for AI Ethics*, the Church has set clear standards. Technology must serve people, not rule them. Systems should lift the poor and protect the vulnerable. Design should strengthen human responsibility, not dull it. True progress builds community and solidarity across peoples and nations.

Serious obstacles remain. The myth that data systems are neutral. Treating persons as data assets. Models that magnify bias. Power locked in a few hands. Unequal access to the benefits of digital tools. Meeting these problems takes more than better code. We need steady moral reflection. Broad, honest conversation with those affected. Firm and fair regulation. Real transparency where possible. Shared commitment from engineers, companies, governments, faith communities, and civil society to put the common good ahead of private gain. When AI work is grounded in human dignity and the common good, as Catholic Social Teaching urges, technology can help heal divides instead of widening them. This is the path to a more just, humane, and inclusive world.

Finally, as the discussion on AI continues to evolve, the principles of Catholic Social Teachings will remain a guide towards building ethical technology. This will help to handle issues around ethical AI, ensuring that we commit to fairness, equal opportunities, and always protecting everyone's basic worth. Catholic Social will continue to help us combine deep moral thinking with new ideas in technology, making sure tech truly benefits everyone and serves the common good (Benedict XVI, 2009).

7.0 References

Note: References include sources cited in the original text and additional key sources relevant to the expanded discussion. Ensure all in-text citations match entries here.

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